

**FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(2019 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(Multiple Choice Questions for SDE Candidates)

1. Let P : Mathematics is interesting, Q : You should not learn it. Then 'Mathematics is interesting and you should learn it' is best represented by :

(A) $\neg P \vee \neg Q$.

(B) $P \wedge \neg Q$.

(C) $P \vee Q$.

(D) $P \wedge Q$.

2. The compound statement $A \rightarrow (A \rightarrow B)$ is false, then the truth values of A, B are respectively.

(A) T, T.

(B) F, T.

(C) T, F.

(D) F, F.

3. The compound propositions p and q are called logically equivalent if _____ is a tautology.

(A) $p \leftrightarrow q$.

(B) $p \rightarrow q$.

(C) $\neg(p \vee q)$.

(D) $\neg p \vee \neg q$.

4. Let $P(x)$ denote the statement " $x > 7$ ". Which of these have truth value true ?

(A) $P(0)$.

(B) $P(4)$.

(C) $P(6)$.

(D) $P(9)$.

5. $p \vee q$ is logically equivalent to :

(A) $\neg q \rightarrow \neg p$.

(B) $q \rightarrow p$.

(C) $\neg p \rightarrow \neg q$.

(D) $\neg p \rightarrow q$.

6. The value of $155 \bmod 9$ is :

(A) 0.

(B) 1.

(C) 2.

(D) 3.

7. If a and b are relatively prime then :
- (A) $a \mid b$. (B) $b \mid a$.
- (C) $\gcd(a, b) = 1$. (D) $\text{lcm}(a, b) = 1$.
8. Let $\gcd(a, b) = d$, then :
- (A) $d = ax + by$, for some x and y . (B) $a \mid d$.
- (C) $b \mid d$. (D) $d = a + b$.
9. The number '1' is :
- (A) Prime number. (B) Composite number.
- (C) Neither Prime nor Composite. (D) None of the mentioned.
10. Difference of two distinct prime numbers is ?
- (A) Odd and prime. (B) Even and composite.
- (C) None of the mentioned.
11. If a, b are integers such that $a > b$ then $\text{lcm}(a, b)$ lies in :
- (A) $a > \text{lcm}(a, b) > b$. (B) $a > b > \text{lcm}(a, b)$.
- (C) $\text{lcm}(a, b) \geq a > b$. (D) None of the mentioned.
12. $(1001111)_2 = \underline{\hspace{2cm}}$.
- (A) 79. (B) 89.
- (C) 69. (D) 99.
13. The linear Diophantine equation $ax + by = c$ has a solution if and only if :
- (A) $\gcd(a, c) \mid b$. (B) $\gcd(a, b) \mid c$.
- (C) $\gcd(c, b) \mid a$. (D) $c \mid \gcd(a, b)$.

14. A composite number n for which $a^n \equiv a \pmod{n}$ is called :
- (A) A pseudoprime. (B) A prime.
- (C) A pseudoprime to the base a . (D) An absolute pseudoprime.
15. If $p, q_1, q_2, \dots, q_n$ are all primes and $p \mid q_1 q_2 \dots q_n$, then :
- (A) $p = q_k$ for some k . (B) $p = 2$.
- (C) $q_k = 2$ for some k . (D) $p \mid q_k$ for some k .
16. If P_n is the n^{th} prime number, then :
- (A) $P_n = n + 1$. (B) $P_n \leq 2^2$.
- (C) $P_n = n! + 1$.
17. If $a \equiv b \pmod{n}$, then :
- (A) a and b leave the same non-negative remainder when divided by n .
- (B) a and b leave the different non-negative remainder when divided by n .
- (C) a and b need not leave the same non-negative remainder when divided by n .
18. If a is an odd integer, then $a^2 \equiv \underline{\hspace{1cm}} \pmod{8}$:
- (A) 1. (B) 2.
- (C) 3. (D) 4.
19. If $a \equiv b \pmod{n}$ and a is a solution of $P(x) \equiv 0 \pmod{n}$, then :
- (A) b is also a solution. (B) b need not be a solution.
- (C) 0 is a solution.
20. The linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if :
- (A) $b = 1$. (B) $b = 0$.
- (C) $d \mid b$ where $d = \gcd(a, n)$.

FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION, NOVEMBER 2020**Mathematics****MTS 1B 01—BASIC LOGIC AND NUMBER THEORY**

(2020 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Find the truth table for the *Disjunction* of two propositions.
2. Define a proposition. Give an example.
3. Define Valid and Invalid Arguments.
4. Show that the proposition $P(0)$ is true where $P(n)$ is the propositional function "If $n > 1$ then $n^2 > n$ ".
5. Prove that every non-empty set of non-negative integers has a least element.
6. Find the quotient q and the remainder r when :
 - (i) 207 is divided by 15.
 - (ii) -23 is divided by 5.
7. Prove that 2 and 3 are the only two consecutive integers that are primes.
8. State and prove Handshake Problem.
9. Define the Fibonacci sequence and write the first four Fibonacci Numbers and Lucas numbers.
10. Using the Euclidean algorithm, express $(4076, 1024)$ as a linear combination of 4076 and 1024.
11. Find the number of trailing zeros in $234!$.
12. Find the largest power of 2 that divides $109!$.
13. Find all solutions of the congruence $9x \equiv 21 \pmod{30}$.
14. If $2p + 1$ is a prime number, prove that $(p)^2 + (-1)^r$ is divisible by $2p + 1$.
15. Find the remainder obtained when 5^{38} is divided by 11.

(10 × 3 = 30 marks)

Turn over

Section B

Answer at least five questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 30.

16. Give a proof by contradiction of the theorem "if n^2 is even, then n is even."
17. Write any 5 inference rules.
18. Prove that there is no positive integer between 1 and 2.
19. Obtain an explicit formula corresponding to the recursive relation :

$$h(n) = h(n-1) + (n-1), n \geq 2.$$

20. Let $(a, b) = d$. Then prove that $(a, a-b) = d$.
21. Explain Jigsaw Puzzle.
22. Find the remainder obtained upon dividing the sum
 $1! + 2! + 3! + 4! + \dots + 99! + 100!$ by 12.
23. Let p be a prime and a any integer such that p does not divide a . Then prove that the solution of the linear congruence $ax \equiv b \pmod{p}$ is given by $x \equiv a^{p-2}b \pmod{p}$.

(5 × 6 = 30 marks)

Section C

Answer any two questions.

Each question carries 10 marks.

24. (a) Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.
 (b) Prove the implication "If n is an integer not divisible by 3, then $n^2 \equiv 1 \pmod{3}$."
25. (a) Prove that a palindrome with an even number of digits is divisible by 11.
 (b) Let a and b be any positive integers, and r the remainder, when a is divided by b . Then prove that $\gcd(a, b) = \gcd(b, r)$.
26. (a) Prove that the gcd of the positive integers a and b is a linear combination of a and b .
 (b) State Duncan's identity. Using recursion, evaluate (18, 30, 60, 75, 132).
27. (a) Prove that no prime of the form $4n + 3$ can be expressed as the sum of two squares.
 (b) Prove that the linear congruence $ax \equiv b \pmod{m}$ is solvable if and only if $d|b$, where $d = (a, m)$ and if $d|b$, then it has d incongruent solutions.

(2 × 10 = 20 marks)

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Mathematics

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

Answer any number of questions.

Each question carries 2 marks.

Maximum 25 marks.

1. Define what is meant by *disjunction* of two statements. Construct the disjunction of the following statements :

Statement 1) Vinod watches cinemas during holidays .

Statement 2) Vinod enjoys poetry during holidays.
2. What is meant by the *converse* of an implication ? Give an example.
3. What is the difference between *tautology* and a *contradiction* ?
4. Show that if $c|a$ and $c|b$, then $c|\alpha a + \beta b$ for any integers α, β .
5. Express $(1092)_{10}$ in base 8.
6. Define the GCD of integers a, b . When do we say that they are relatively prime ?
7. Find the canonical decomposition of 1980.
8. Find $(252, 350)$ and hence find $[252, 360]$.
9. If $a \equiv 5 \pmod{25}$ give four possibilities of a .
10. Which of the following is a complete set of residues modulo 8?
 $\{1, 2, 4, 5, 7, 8, 11, 14\}$ or $\{1, 2, 4, 5, 7, 8, 19, 24\}$ or both ? Why ?
11. Find the inverse of 7 modulo 50.

Turn over

12. Evaluate $\phi(35), \phi(48)$ directly without using formula.
13. State Fermat's Little Theorem. Prove it.
14. Define the function τ . Evaluate $\tau(19)$ and $\tau(23)$.
15. Compute $\phi(666)$ and $\phi(1976)$ using canonical decomposition.

Section B

Answer any number of questions.

Each question carries 5 marks.

Maximum 35 marks.

16. Test the validity of the following argument :

A_1 There are more residents in New Delhi than there are hairs in the head of any resident.

A_2 No resident is totally bald.

Hence At least two residents must have the same number of hairs on their heads.

17. State the Inclusion-Exclusion Principle. Use it to find the number of positive integers ≤ 2076 that are divisible by neither 4 nor 5.
18. Define Fermat numbers. Derive a recurrence formula for the n^{th} Fermat number f_n .
19. Let a and b be any positive integers, and r the remainder, when a is divided by b . Prove that $(a, b) = (b, r)$.
20. Find the general solution to the LDE $12x + 20y = 28$.
21. Prove that no integer of the form $8n + 7$ can be expressed as a sum of three squares.
22. Let p be a prime and a any positive integer. Prove that $a^p \equiv a \pmod{p}$. Does this result hold for some non-prime integer p ? Justify.
23. Prove that if n is an odd pseudoprime, then $N = 2^n - 1$ is also an odd pseudoprime.

Section C

*Answer any **two** questions.*

Each question carries 10 marks.

Maximum 20 marks.

24. (a) Prove directly that the product of any even integer and any odd integer is even.
- (b) Prove by cases that for any integer n , $n^2 + n$ is an even integer.
- (c) Prove by contradiction that $\sqrt{5}$ is an irrational number.
25. (a) Let p be a prime and $p \mid a_1 a_2 \dots a_n$, where $a_1, a_2, \dots, a_n$ are positive integers. Prove that $p \mid a_i$ for some i , where $1 \leq i \leq n$.
- (b) State and prove the Fundamental Theorem of Arithmetic.
26. Prove that the linear congruence $ax \equiv b \pmod{m}$ is solvable if and only if $d \mid b$, where $d = (a, m)$.
Also, prove that if $d \mid b$, then it has d incongruent solutions.
27. Prove that the function ϕ is multiplicative. Use it to evaluate $\phi(221)$ and $\phi(6125)$.